Short Term Interest Rate Innovations Across Monetary Regimes

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Status: Documented reconstruction. Every estimate reported here is produced from source-compatible reconstructions rather than from the original article's own input files, which are not distributed. No result carries an exact-replication label, and the reasons are set out in Section 4.

Abstract

This paper asks whether short-term interest-rate innovations help explain the cross section of stock-market anomaly returns, and whether that relation is stable after the publication sample and across monetary regimes. We rebuild the design of Maio and Santa-Clara (2017) from independently acquired sources, holding every threshold and decision rule fixed in advance, and separate documented reconstruction from new extension evidence throughout. On the 1972–2013 baseline the reconstruction recovers the published pattern: the short-rate innovation carries a negative price of risk that is significant under the Shanken correction, and it cuts cross-sectional pricing errors against the market model by roughly a half. Three pre-registered findings qualify that. First, the incremental-pricing standard registered in advance is not met on the headline asset set, and every traded multi-factor comparator prices the same seventy portfolios more accurately than the short-rate model does. Second, the relation does not survive the publication sample: refitted post-2013 estimates reverse sign for five of seven anomaly families, and we show this is not a data-vintage artifact and not, by the registered identification gate, a weak-factor artifact. Third, exposures are regime dependent, with the pooled interaction test rejecting stability decisively, while the regime-specific equivalence tests are too imprecise at feasible regime lengths to certify either equivalence or difference for any individual portfolio. A power analysis shows this imprecision is a property of the estimator at those sample sizes rather than a finding about the periods themselves.

1 Introduction

Short-term interest rates are plausible state variables in intertemporal asset-pricing models because they summarize investment opportunities that matter for both discount rates and hedging demands. If portfolios formed on accounting or return-based anomalies have different exposures to those state variables, a rate innovation may help organize average excess returns beyond the market factor. The economic question is not only whether such a factor prices anomaly returns in one sample, but whether the relation survives changes in monetary regimes and source definitions.

Maio and Santa-Clara (2017) provide the starting point. Their article studies short-term interest-rate innovations and stock-market anomalies, and this paper uses that design as the baseline object to be reproduced before any extension is interpreted. The strict baseline window is January 1972 through December 2013. The extension window begins in January 2014 and is evaluated separately from the replication verdict.

The paper has three aims. First, it rebuilds the original baseline in a way that distinguishes exact replication from documented reconstruction. Second, it asks whether the short-rate factor is stable after the publication sample and across monetary regimes. Third, it evaluates whether any short-rate pricing evidence is robust to weak-factor diagnostics and comparator-model discipline. The high-frequency policy-information decomposition is kept as an appendix design until usable event data and monthly component artifacts exist.

Four findings follow. On the baseline window the reconstruction recovers the published qualitative result: the short-rate innovation earns a negative price of risk of -0.6985 per month with a Shanken t of -2.86, and it roughly halves cross-sectional pricing errors relative to the market model. Against that, the incremental-pricing standard we registered in advance is not met on the joint asset set, and every traded multi-factor comparator we examine prices the same portfolios with smaller errors than the short-rate model. The relation then fails to extend past the publication sample, in a way we can attribute to the passage of time rather than to data revisions or to weak identification. Finally, short-rate exposures differ across monetary regimes, though the sample lengths available inside regimes are too short for the registered equivalence standard to resolve the size of the difference.

Two of these findings are negative, and we regard that as the substantive contribution rather than as a failed replication. The baseline result reproduces; what does not survive is its extension. Because every threshold, comparator, and decision rule was fixed before the corresponding estimate existed, the negative findings are not the product of specification search, and we report each registered gate whether it passed or failed.

2 Related literature

The first literature stream motivates the state variable. The ICAPM shows why assets can earn premia for covariance with variables that forecast future investment opportunities (Merton, 1973). Interest-rate applications of this idea are directly relevant because Lioui and Maio (2014) model interest-rate risk as a priced source of cross-sectional equity returns, and Maio and Santa-Clara (2017) apply short-rate innovations to anomaly portfolios.

The second stream concerns anomaly pricing and model comparison. The baseline article compares a short-rate ICAPM with standard factor models, so the extension must be evaluated against asset-pricing benchmarks rather than against the CAPM alone. Fama and French (2015) add profitability and investment factors to the Fama-French framework, while Hou et al. (2015) propose an investment approach that summarizes many anomalies with market, size, investment, and profitability factors. These benchmarks make the extension question sharper: does the short-rate factor add robust pricing content once comparator models, asset intersections, and portfolio definitions are held fixed?

The third stream concerns inference. Two-pass cross-sectional tests are exposed to generated-regressor and weak-factor problems. Fama and MacBeth (1973) provide the classic cross-sectional testing framework, Shanken (1992) derives corrections for beta-estimation error, and Kan and Zhang (1999) show that useless factors can appear priced in two-pass tests. Lewellen et al. (2010) argue that high cross-sectional fit and small pricing errors can be misleading when test assets share a strong characteristic structure. This paper therefore treats standardized exposure dispersion, rank, spanning, and influence diagnostics as conditions for interpretation rather than optional robustness checks.

The fourth stream concerns monetary regimes and high-frequency identification. Bernanke and Kuttner (2005) measure equity-market responses to unanticipated Federal Reserve policy actions, while Jarociński and Karadi (2020) separate policy and central-bank-information components using high-frequency announcement comovement. Swanson and Williams (2014) show why lower-bound episodes can change the informativeness of short-term rates for the broader term structure. Those papers motivate separating the aggregate AR rate innovation from identified announcement components and from regime-specific pricing evidence. Structural-change methods such as Bai and Perron (1998) provide a way to distinguish predeclared regime boundaries from estimated breaks.

The final stream is data vintage discipline. Official factor libraries publish current data, dated historical archives, and selected change notes rather than a complete revision history, so the retrieval vintage of a source has to be recorded before an estimate is labelled an exact replication or entered into a post-publication comparison (French, 2026). This paper therefore adds value beyond the original article by freezing a table-level replication target before asking whether the short-rate evidence generalizes over time and across regimes.

3 Original framework and economic mechanism

The original framework is a discrete-time ICAPM application in which the market excess return is paired with an innovation in a short-term interest-rate state variable. The article verifies two state variables: the federal funds rate and the three-month Treasury-bill rate. It motivates both series as proxies for investment opportunities and constructs innovations from first-order autoregressions.

For the federal-funds version, the article reports the state-variable law of motion as

$$FFR_{t+1} = 0.000 + 0.991FFR_t + u_{t+1}^{FFR}, \quad R^2 = 0.98,$$

which this manuscript indexes for the return month as

$$u_t^{FFR} = FFR_t - \hat{a} - \hat{\rho}FFR_{t-1}.$$

This convention fixes the information timing: the return regression for month t uses the innovation from the same realized month and never relies on an implicit shift. The article reports the analogous Treasury-bill equation as

$$TB_{t+1} = 0.000 + 0.992TB_t + u_{t+1}^{TB}, \quad R^2 = 0.98.$$

For test asset i, excess return $R_{i,t}^e$, market factor $R_{M,t}$, and federal-funds innovation u_t^{FFR}, the verified first-pass time-series regression is

$$R_{i,t}^e = \delta_i + \beta_{i,M}R_{M,t} + \beta_{i,FFR}u_t^{FFR} + \varepsilon_{i,t}.$$

The benchmark second-pass regression is a single OLS cross-sectional regression of full-sample average excess portfolio returns on full-sample first-pass betas, without an intercept:

$$\bar{R}_i^e = \hat{\beta}_{i,M}\lambda_M + \hat{\beta}_{i,FFR}\lambda_{FFR} + \alpha_i^{CS}.$$

Here δ_i is the time-series intercept, α_i^{CS} is the cross-sectional pricing error, and λ_{FFR} is the federal-funds innovation risk price. This benchmark is not described as repeated monthly Fama–MacBeth estimation unless a separate supplement target or robustness artifact explicitly implements repeated cross sections. The no-intercept restriction is part of the article's benchmark economic test, not a default regression convention.

The economic mechanism is an ICAPM-style hedging argument. A rate innovation can matter if it captures news about future investment opportunities and if anomaly portfolios differ in their exposure to that news. That statement is weaker than a monetary-policy interpretation. An autoregressive residual may combine policy news, information about the macroeconomy, data revisions, and persistence in the short-rate series. The paper therefore treats the baseline factor as a rate innovation until a separate high-frequency design justifies narrower language.

The article's primary test assets are value-weighted decile portfolios sorted on book-to-market, equity duration, earnings-to-price, long-term reversal, investment-to-assets, property-plant-and-equipment investment, and inventory growth. It also studies the joint system of all seven anomaly families and uses equal-weighted portfolios and additional appendix designs as robustness checks. The benchmark inference reports Shanken-corrected risk-price statistics, bootstrap p-values, a joint pricing-error statistic, and cross-sectional fit metrics.

The mechanism is not one channel. Short-rate innovations may proxy for discount-rate news, cash-flow news, financing-cost changes, working-capital conditions, monetary-policy information, or intertemporal hedging demand. The equity-duration channel is the clearest family-level mechanism because a rate innovation changes the discounting of cash flows at different horizons. For the other anomaly families, the paper predeclares exposure comparisons but does not force a directional sign unless the article-level result and generated portfolio definition support it.

The meaning of a numerical rate innovation can also change across regimes. Conventional positive-rate policy, lower-bound periods, asset-purchase and forward-guidance episodes, and inflation-driven tightening can attach different information to the same measured short-rate movement. The baseline rate residual therefore remains an aggregate state-variable innovation; policy and information language is reserved for the optional announcement-based appendix design.

Some implementation details remain ambiguous at the source-file level. The article identifies the Federal Reserve Bank of St. Louis, Kenneth French's data library, Robert Stambaugh's web page, and Lu Zhang as sources, but the exact download files and archive vintages still have to be frozen before the manuscript can claim exact computational replication.

4 Data and replication design

The data design separates exact inputs, approximate reconstructions, and extension inputs. Exact inputs require a verified article definition, source, vintage, transformation, and estimator role. Approximate reconstructions are documented public or author-data substitutes and are never labelled as exact replication. Extension inputs support post-publication, regime, and high-frequency analyses after their source terms and transformations are frozen.

The baseline sample runs from 1972-01 through 2013-12. The temporal extension begins in 2014-01, with revised-history checks kept separate from the baseline verdict. Every source must record access status, redistribution status, and its role in strict replication before it can support a claim. Exact unavailable inputs lead to a missing-input classification rather than an empirical contradiction.

The short-rate registry fixes the documented reconstruction series before the AR model is estimated. The federal-funds reconstruction uses FRED series FEDFUNDS, an effective federal funds rate, monthly provider average, annualized percent units, no seasonal adjustment, no imputation, and a frozen retrieval vintage. The Treasury-bill reconstruction uses TB3MS as the monthly secondary-market candidate and treats DTB3 only as a sensitivity candidate, because

Table 1: Replication target and economic hypotheses

Claim	Role	Primary test	Interpretation rule
R1a–R1f	Replication targets	Layer-specific article statistics within tolerances	Numerical recovery only
H1	Incremental pricing content	Improvement versus the ex ante CAPM comparator	Economic thresholds required against CAPM; the strongest observed comparator is a secondary adversarial check
H2	Temporal stability	Post-publication compatibility	Sign and magnitude matter
H3	Regime stability	Separate beta, fitted-premium, and error tests	Non-rejection is insufficient
H4a	Cross-sectional identification strength	Beta rank, standardized exposure dispersion, and the numerical spanning criterion	Weak identification blocks interpretation
H4b	Influence stability	Leave-one-family refits and standardized influence	One family or one portfolio may not drive the conclusion
H4c	Fitted-premium precision	Joint-bootstrap interval for the rate-attributable fitted premium	An interval spanning both economic directions is uninterpretable

Source: Authors' replication protocol.

neither the article nor the supplement names a daily source or a daily-to-monthly aggregation rule. The monthly aggregation of both series is verified rather than assumed: each provider monthly value equals the corresponding daily mean rounded to the published precision in every complete month, and two competing aggregation rules are rejected. None of these rate choices is eligible for an exact replication label, because the article names a provider without a series code or an archive vintage. Source identity, not numerical proximity, decides the replication mode.

Portfolio continuity is the largest post-publication constraint. All seven registered anomaly families are available from the author source the article names, but only in a post-publication vintage that runs to 2025-12; no publication-era vintage of those files is publicly recoverable. The extension therefore proceeds as a documented reconstruction on the author's own updated panels, and the portfolio sample rather than the rate and factor data sets the binding extension endpoint. Security-level reconstruction from CRSP and Compustat remains the only route to an exact continuation; database access is not assumed and redistribution of licensed extracts is prohibited, so that route is not taken. Portfolio catalogues from other providers are never substituted for a named author source, and no acquired portfolio supports an exact-replication claim unless source compatibility is independently verified.

Short-rate series are stored in their source units and transformed only in named innovation steps. Market and portfolio returns are aligned to the canonical monthly panel before excess returns, factor construction, or model comparison. Missing returns and shock observations are not forward filled. Comparator models use identical asset-date intersections within each comparison, and delisting or survivorship treatment is recorded from verified source documentation rather than inferred from the data file alone.

The replication target is table-level. Each published table or appendix table is mapped to a

Table 2: Data sources and sample-construction status

Variable or family	Source	Frequency	Transformation	Status
Article and supplement	Publisher files	Document	Target extraction	Exact local access
Federal funds rate	Federal Reserve Bank of St. Louis	Monthly	FEDFUNDS documented reconstruction; first-order innovation	Acquired and frozen; aggregation verified; not exact
Treasury-bill rate	Federal Reserve Bank of St. Louis	Monthly	TB3MS documented reconstruction; DTB3 sensitivity only	Acquired and frozen; aggregation verified; not exact
Market and risk-free returns	Kenneth French Data Library	Monthly	Excess-return factors	Acquired at a publication-era and a current vintage; not exact
Anomaly decile portfolios	Lu Zhang and cited sources	Monthly	Excess returns after alignment	Acquired from the named source at a post-publication vintage; equal-weighted variants unavailable
Comparator factors	French, Stambaugh, and q-factor sources	Monthly	Common-intersection factors	All registered comparator inputs acquired and frozen; the liquidity column and scale are identified empirically and its published extremes are reproduced by no recoverable vintage
Announcement components	Documented research dataset	Event	Monthly aggregation	Extension input pending

No input is exact, because the article names providers and people rather than files. Source: `research/data_access_matrix.csv`.

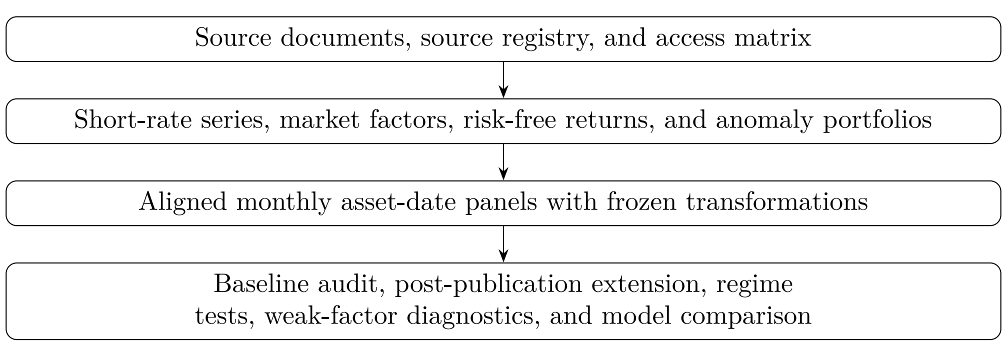

Figure 1: Data flow from sources to manuscript evidence
Source: Authors' replication protocol.

source location, model, estimator, tolerance rule, and status. A generated statistic can be labelled reproduced only after the same definition and estimator are linked to an auditable output. Nearby public data may be useful for documented reconstruction, but they cannot change the strict replication label. The replication target is decomposed into short-rate innovations, first-pass betas, cross-sectional risk prices, pricing errors and fit, comparator-model results, and supplementary robustness targets.

5 Econometric methods

The baseline empirical design starts with the article's first-order short-rate innovation equations. Alternative autoregressive orders, alternative rate definitions, local-level filters, and announcement-surprise measures are robustness or extension specifications, not replacements for the strict baseline.

First-pass time-series regressions estimate portfolio exposures to the market excess return and the selected short-rate innovation. Their standard errors are heteroskedasticity and autocorrelation consistent, computed with the Newey and West (1987) covariance estimator under an automatic monthly lag rule that depends only on the number of observations, so no lag length is chosen after seeing an estimate. The notation separates the time-series intercept δ_i, factor loadings β_i, residuals $\varepsilon_{i,t}$, cross-sectional pricing errors α_i^{CS}, and risk prices λ. The second pass is the article's verified no-intercept OLS regression of average returns on estimated betas unless a supplement target explicitly calls for an unrestricted zero-beta-rate variant, GLS estimator, or repeated monthly cross-sectional estimator. Shanken corrections apply to the reported full-sample risk-price estimator in the benchmark design; bootstrap procedures are labelled by resampling unit before any p-value is interpreted.

Model comparison reports pricing-error loss and fit on common asset-date intersections. The registered outcomes are root mean squared pricing error, mean absolute pricing error, maximum absolute pricing error, cross-sectional fit, and joint pricing-error tests when the estimator supports them. Statistical significance is not enough: every supported result must also clear the registered economic-magnitude criterion. For H1, materiality is defined first against the ex ante CAPM comparator on the common intersection. The short-rate model must reduce RMSE and MAE by at least 10 percent and reduce the maximum absolute pricing error by at least 0.25 monthly percentage points relative to CAPM. The strongest observed registered non-short-rate comparator is retained only as a secondary adversarial analysis with model-selection uncertainty.

Weak-factor diagnostics evaluate whether short-rate betas contain enough cross-sectional variation to support a pricing interpretation. They are split into three separately classified confirmatory claims rather than one composite gate, because identification strength, influence stability, and estimation precision fail for different reasons and carry different remedies. Raw beta dispersion, singular values, and condition numbers may be reported descriptively, but they are not separate confirmatory failure rules. A short-rate risk price that fails any of the three is interpreted as weakly identified even when fit metrics appear favorable.

H4a is cross-sectional identification strength. It fails when the beta matrix loses rank, when standardized rate-exposure dispersion is less than 10 percent of standardized market-exposure dispersion, or when the numerical spanning criterion fails. Spanning is decided mechanically: the short-rate innovation is regressed by OLS on a constant and the registered non-short-rate comparator factors available on the same intersection, and the gate fails when the spanning R^2 exceeds 0.90, equivalently when the residual standard deviation falls below $\sqrt{0.10}$ of the raw factor standard deviation. The R^2 form is authoritative and the residual cutoff is the exact square root rather than a decimal rounded below it, so the two forms agree on every boundary case. Both statistics are scale free. H4b is influence stability, which fails when a leave-one-family refit reverses the sign or materiality of the rate-attributable fitted-premium spread or when a single portfolio reaches a standardized influence of one on the rate risk price. H4c is fitted-premium precision, which fails when the 95 percent joint-bootstrap interval for the fitted-premium spread contains both $+0.25$ and -0.25 monthly percentage points.

Temporal and regime methods are designed as extensions. The post-publication analysis has frozen-parameter and refitted variants, with no-lookahead rules for factor construction, portfolio membership, model vintage, and sample intersections. Regime analysis separates four questions: stability of the rate innovation construction, stability of portfolio betas, stability of cross-sectional risk prices, and stability of pricing errors. Pooled time-series interactions test beta stability; regime-specific or locally estimated second-pass models test risk-price stability. Stability is not inferred from a failure to reject equality.

Regime risk-price stability is evaluated through the rate-attributable fitted premium $\pi_{i,s} = \beta_{i,FFR,s}\lambda_{FFR,s}$, not through an absolute bound on $\lambda_{FFR,s}$. This object is invariant to rescaling the short-rate innovation. The predeclared equivalence bounds require fitted-premium changes no larger than 0.25 monthly percentage points, fitted-premium dispersion changes no larger than 25 percent, RMSE deterioration no larger than 10 percent, maximum pricing-error deterioration no larger than 0.25 monthly percentage points, and selected-fit deterioration no larger than 0.10 under the fit contract.

Cross-sectional fit is mechanical rather than discretionary. The primary fit statistic is the article's exact definition if verified; otherwise the fallback is a no-intercept pricing-error pseudo-R^2, $1 - \sum_i(\alpha_i^{CS})^2 / \sum_i(\bar{R}_i^e)^2$, reported only when the denominator is positive and all models use the same asset-date intersection.

Inference uses Shanken-adjusted covariance for the benchmark full-sample second-pass risk-price estimator. Fitted-premium uncertainty uses a joint moving-block bootstrap with 10,000 repetitions, a Politis-White block-length selector, and a fixed 12-month fallback under declared selector failures. The stochastic variables resampled jointly by calendar month are the market factor, the comparator factors, the portfolio returns, and the short-rate level series. Regime labels are not among them: a label is a deterministic calendar function of the observation month, so each resampled month carries its own frozen label as a fixed attribute. Resampling labels would fabricate uncertainty about a known policy calendar. Each draw re-estimates the rate innovation, first-pass betas, second-pass risk prices, fitted premia, pricing errors, fit metrics, and equivalence statistics. Equivalence claims use one rule: standard two one-sided tests at the 5 percent level, implemented as inclusion of the two-sided 90 percent bootstrap interval inside the declared bound. The stricter 95 percent interval-inclusion variant is reported only as a labelled sensitivity column and never as the confirmatory test.

The primary regime registry contains six regimes: conventional positive-rate policy, effective lower bound and first QE, first normalisation, pandemic lower-bound and QE, inflation-driven tightening, and post-tightening easing and normalisation. The post-2022 period is split because a rapid tightening cycle and the subsequent reduction from the cycle peak attach different information to the same measured rate movement; the split boundary is the first policy reduction on 18 September 2024. The primary transition rule assigns a regime to the first calendar month that lies entirely under the new policy stance, so the month containing the policy action stays with the outgoing regime. The whole-transition-month rule, which assigns the action month to the incoming regime, is preserved as a declared sensitivity analysis and never decides a confirmatory classification. Boundary-shift checks remain plus or minus three months. The entire pre-lower-bound sample is defensible as the confirmatory article-era regime because the federal funds rate is the primary short-rate policy instrument away from the lower bound; finer pre-lower-bound breaks remain exploratory and cannot replace the confirmatory calendar regime.

Regime-estimation eligibility is frozen numerically before any regime estimate is produced and is not relaxed for regimes that turn out to be short. A regime shorter than 36 months supports no regime-specific estimation and enters only pooled regime-interaction models. A regime of 36 to 59 months supports regime-specific first-pass betas under a short-sample flag but no standalone second pass. A standalone regime-specific second pass additionally requires at least 60 months, at least 10 test assets, and a beta matrix whose rank equals the number of priced factors. Under these frozen rules only the conventional and lower-bound regimes carry full eligibility, and the three post-2020 regimes are restricted to pooled interactions; a predeclared combination of the two post-2022 regimes is available as a first-pass sensitivity.

Table 3: Predeclared monetary regimes

ID	Regime	Start	End	Obs.	Primary instrument	Eligibility
`conventional_pre_elb`	Conventional positive-rate policy	1972-01	2008-12	444	Federal funds rate	First pass and standalone second pass
`elb_qe`	Effective lower bound and first QE	2009-01	2015-12	84	Target range and asset purchases	First pass and standalone second pass
`normalisation`	First normalisation	2016-01	2020-03	51	Target range and balance-sheet normalisation	First pass with short-sample flag; standalone second pass blocked
`pandemic_elb_qe`	Pandemic lower-bound and QE	2020-04	2022-03	24	Lower-bound target range and asset purchases	Below the 36-month floor; pooled interactions only
`inflation_tightening`	Inflation-driven tightening	2022-04	2024-09	30	Federal funds target range	Below the 36-month floor; pooled interactions only
`post_tightening_easing`	Post-tightening easing and normalisation	2024-10	2026-06 registered, 2025-12 effective	21 registered, 15 usable	Federal funds target range	Below the 36-month floor; pooled interactions only

A regime begins in the first calendar month lying entirely under the new policy stance, so the month containing the policy action stays with the outgoing regime; the whole-transition-month rule is a declared sensitivity. Permitted boundary shifts are plus or minus three months. Registered observation counts are those frozen in the regime registry, which ends the final regime at the rate-data freeze month 2026-06. Regime estimation uses the registered anomaly decile portfolios, which end 2025-12, so the last six months of the post-tightening easing window carry no test-asset observations and the usable count for that regime is 15 rather than 21. The declared post-2022 combination is affected in the same way and has 45 usable months rather than 51. The registry rows are not restated here; the effective endpoint is the binding constraint on every regime estimate. Source: `research/regime_registry.csv`, `research/anomaly_portfolio_availability.csv`, `configs/extensions.yaml`.

The high-frequency decomposition is optional rather than part of the main contribution. If included, it starts from high-frequency components constructed under an explicit announcement-based identification design after source terms and component definitions are frozen. The monthly aggregation can create a sparse factor with many no-event months, so the component factors must pass their own weak-factor diagnostics before entering the main argument.

Out-of-sample evaluation defines the forecast origin, training window, refit schedule, benchmark models, loss functions, and interpretation before results are inspected. It is an appendix robustness exercise unless it produces evidence strong enough to motivate a separate contribution. Multiplicity adjustments are predeclared within replication, baseline pricing, temporal extension, regime stability, and weak-factor families.

Table 4: Method-to-claim map

Claim	Primary method	Decision rule
R1a–R1f	Layer-specific comparison to registered article statistics	Recovery within declared tolerances
H1	No-intercept two-pass pricing and model comparison	Must beat ex ante CAPM by materiality thresholds; strongest observed comparator is secondary
H2	Frozen and refitted post-publication evaluation	Sign, magnitude, and loss compatibility must hold
H3	Separate beta, fitted-premium, and pricing-error stability tests	Estimates must remain inside numerical bounds under 5 percent TOST
H4a	Beta rank, standardized exposure dispersion, and spanning regression	All three identification gates must pass
H4b	Leave-one-family refits and standardized influence diagnostics	No sign reversal, no materiality loss, and no dominant portfolio
H4c	Joint moving-block bootstrap of the rate-attributable fitted premium	The interval may not span both economic directions

Source: Authors' replication protocol.

6 Baseline replication

6.1 The short-rate innovation

The first layer to reproduce is the state variable itself. Estimating the article's autoregression on independently acquired federal funds and Treasury-bill series over the baseline window gives the coefficients in Table 5. The persistence parameters agree with the published values at printed precision for both series, and the fit statistics agree to two decimals.

One conversion is required before the intercepts can be compared. The article prints its autoregressive intercept as 0.000 while printing the innovation descriptives of its Table 1 in percent, so the intercept is stated in decimal rate units and must be divided by one hundred before comparison. Applying that conversion, both reconstructed intercepts round to the published 0.000; without it, neither does. The slope, both t-ratios, and the coefficient of determination are scale invariant and need no conversion.

Table 5: Short-rate innovation reconstruction, 1972-01 to 2013-12, under the recovered pre-window-lag timing convention. Published values are those printed by Maio and Santa-Clara (2017). The intercept is in decimal rate units and the innovation standard deviation in annualized percentage points.

Series	Intercept	Slope	Slope t	R^2	Innovation s.d.
Federal funds, reconstructed	0.000463	0.9905	147.27	0.9774	0.5855
Federal funds, published	0.000	0.991	—	0.98	0.59
Treasury bill, reconstructed	0.000354	0.9916	153.18	0.9791	0.4855
Treasury bill, published	0.000	0.992	—	0.98	0.49

The timing convention is not stated in the article and had to be recovered. Two variants are admissible: regressing the first window month on the level of the month preceding the window, or beginning the regression one month inside the window. Only the first reproduces the published slopes, t-ratios, and fit statistics, and it is the variant used throughout. Under it the federal funds reconstruction matches nine of ten registered Table 1 statistics and the Treasury-bill reconstruction matches ten of ten.

Neither series earns an exact-replication label. The article names its providers without giving series codes or archive vintages, so no reconstruction of the rate is eligible for that label at any recovery rate, and both are classified `approximately_reproduced_under_documented_reconstruction`.

6.2 Cross-sectional pricing

Table 6 reports the second pass for all eight registered models on the joint seventy-portfolio system over the 504 baseline months.

Table 6: Baseline cross-sectional pricing, joint seventy-portfolio system, 1972-01 to 2013-12, 504 months. Risk prices are monthly percentage points with Shanken t-ratios beneath. Fit is the article's centred cross-sectional variance ratio. All rows carry the label `documented_reconstruction`.

Model	λ_{RM}	λ_{rate}	RMSE	MAE	$\max\|\alpha\|$	Fit	χ^2 (p)
CAPM	0.648	—	0.1893	0.1466	0.4752	−0.613	99.57 (0.009)
	(3.12)						
Market + funds-rate	0.601	−0.698	0.1004	0.0816	0.2388	0.544	35.65 (1.000)
	(2.87)	(−2.86)					
Market + bill-rate	0.620	−0.482	0.1084	0.0858	0.2781	0.468	46.77 (0.977)
	(2.97)	(−3.02)					
Fama-French three	0.612	—	0.0798	0.0630	0.2409	0.712	82.31 (0.098)
	(2.95)						
Carhart four	0.607	—	0.0720	0.0560	0.2427	0.765	66.82 (0.449)
	(2.93)						
Fama-French five	0.612	—	0.0745	0.0595	0.1743	0.749	78.14 (0.127)
	(2.95)						
q-factor	0.608	—	0.0832	0.0664	0.2164	0.687	81.45 (0.095)
	(2.94)						
Liquidity	0.615	—	0.0796	0.0623	0.2354	0.713	81.77 (0.091)
	(2.97)						

The qualitative published result reproduces. Adding the funds-rate innovation to the market model lowers cross-sectional RMSE from 0.1893 to 0.1004 and raises the article's fit statistic from −0.613 to 0.544, and the rate carries a negative price of risk with a Shanken t of −2.86. The Treasury-bill specification gives the same sign and a comparable t-ratio, so the finding does not hinge on which short rate is used.

Two features of the table qualify that reading, and both work against the short-rate model. First, every traded multi-factor comparator attains a lower RMSE and a higher fit than the short-rate specification: the Carhart model reaches an RMSE of 0.0720 against 0.1004. The short-rate model is a large improvement on the market model alone and is simultaneously the weakest of the multi-factor specifications on these test assets.

Second, the specification test is not evidence in the short-rate model's favour despite its p-value of essentially one. The Shanken multiplier for that model is 2.42, against 1.02 for the market model and between 1.06 and 1.11 for the traded-factor models. The multiplier scales the estimated covariance of the pricing errors, so a model with a large price of risk relative to factor volatility inflates the variance against which its own pricing errors are judged, and the statistic falls. The non-rejection reflects that inflation as much as it reflects small pricing errors, which is why the comparison above rests on RMSE rather than on χ^2.

6.3 What the cell-level audit recovers

Comparing the reconstruction against the published cells gives a recovery rate that varies sharply and informatively by statistic. Of 123 unique published cells, all 123 were compared. Market risk prices recover well, with 20 of 29 cells inside the published rounding and all 29 agreeing in sign. Rate risk prices recover in sign and magnitude but less often at printed precision, with all 16 cells negative as published and a median absolute difference roughly four times that of the market factor. The χ^2 statistic recovers least well, with none of 29 cells inside the published rounding.

That ordering is what the estimator predicts. The specification statistic inverts a seventy-by-seventy pricing-error covariance through a pseudo-inverse, which amplifies small differences in the residual covariance, so a portfolio difference that moves a risk price by 0.003 can move the statistic by a whole unit.

The common source is the portfolio vintage. The anomaly decile panels come from the original author's source but in a later rebuild, and none of the seven families matches its published descriptive row on all five statistics. A cross-sectional estimator applied to slightly different portfolios returns slightly different cells. Nothing in the audit separates that explanation from an estimator discrepancy, and we do not claim otherwise. Accordingly the layers are classified as partially recovered under documented reconstruction rather than as reproduced or contradicted; a contradiction label would require a completed attempt on source-compatible inputs, which these are not.

The baseline analysis uses the strict monthly panel, the registered baseline and comparator system, the article's two-pass estimator with the Shanken correction, and the seventy registered anomaly portfolios. Uncertainty is the Shanken covariance applied to both risk prices and pricing errors; the economic magnitude is a reduction in cross-sectional RMSE of 47.0 percent relative to the market model.

7 Weak-factor and model-comparison diagnostics

Two questions govern whether the baseline result may be interpreted at all. The mechanical one is whether estimated short-rate betas vary enough across assets for the second pass to identify a price of risk. The economic one is whether the factor reduces pricing errors by a material amount against comparator models chosen in advance. The first is answered affirmatively; the second is not.

7.1 The factor is well identified

All three identification gates pass. The beta matrix attains full rank. The standardized rate-exposure dispersion is 0.2540 of the standardized market dispersion against a floor of 0.10, and per family it ranges from 0.1903 to 0.3894, so no family sits near the floor. The spanning regression of the rate innovation on all ten registered non-short-rate comparator factors returns a residual standard-deviation ratio of 0.9716 against a required minimum of $\sqrt{0.10} \approx 0.3162$: the rate innovation is a long way from being spanned by the existing traded-factor set.

Influence is likewise stable. Across the seven leave-one-family refits the rate price of risk moves within $[-0.7519, -0.6777]$ with no sign reversal, and the largest standardized influence of any single portfolio on it is 0.0896 against a bound of one.

Fitted-premium precision passes for every family. The rate-attributable fitted-premium spread between extreme deciles has a bootstrap interval that excludes at least one economic direction in all seven families, so no family is so imprecisely estimated that its sign is uninformative. Intervals come from a moving-block bootstrap with 10,000 repetitions in which each draw resamples calendar months jointly and recomputes the entire chain: the autoregression, all seventy first-pass betas, the no-intercept second pass, and the fitted premia. The block length of six months was selected by the Politis-White rule rather than set by hand.

Taken together these say the short-rate factor is well enough identified, stable enough to test-asset composition, and precisely enough estimated to carry a pricing interpretation. They do not say the interpretation is correct.

7.2 The incremental-pricing standard is not met

The materiality standard was registered before any estimate existed, with the CAPM named in advance as the primary comparator and all three gates required to hold. Table 7 reports them.

Table 7: Incremental-pricing materiality against the ex ante CAPM comparator, joint seventy-portfolio system. All three gates must hold for the claim to be supported; the registered classification is `unsupported`.

Gate	Threshold	CAPM	Market + rate	Result
RMSE reduction	$\geq 10\%$	0.1893	0.1004	pass, 47.0%
MAE reduction	$\geq 10\%$	0.1466	0.0816	pass, 44.3%
$\max \lvert\alpha\rvert$ reduction	≥ 0.25 pp	0.4752	0.2388	**fail**, 0.2364 pp

The registered classification is therefore *unsupported*. The two relative gates clear by roughly a factor of four; the absolute gate fails by 0.0136 monthly percentage points.

That asymmetry deserves to be on the record, because it alone determines the headline classification. The market model's own maximum absolute pricing error on this asset set is 0.4752 monthly percentage points, so demanding an absolute reduction of 0.25 demands cutting that statistic by 52.6 percent. The short-rate model cuts it by 49.7 percent. In relative terms the third gate is a five times stricter bar than the first two, which ask for 10 percent. Nothing in the registered thresholds states that the asymmetry was intended. It was nonetheless fixed before any estimate existed and is applied exactly as fixed; adjusting it now, in the knowledge that it is the binding constraint, would not be defensible. We record it as a design limitation in Section 10 rather than revising it.

Two of the seven families, equity duration and investment to assets, are supported under both the funds-rate and the bill-rate specifications. The other five and the joint system are not, and the absolute gate is the binding one in six of the eight asset sets.

A secondary comparison against the strongest observed non-short-rate comparator, selected *after* observing baseline RMSE and therefore carrying model-selection uncertainty, is unsupported in all sixteen rows with every gate failing. This selection never feeds the choice of primary comparator, and it is reported because omitting it would overstate the short-rate model's standing.

These diagnostics use the strict baseline panel, the registered robustness family, the article's two-pass estimator with bootstrap and influence diagnostics, and the seventy registered anomaly portfolios. Uncertainty is governed by the weak-factor gates and the joint moving-block bootstrap; the economic magnitude is a shortfall of 0.0136 monthly percentage points on the single binding materiality gate.

8 Post-publication evidence

The baseline endpoint remains 2013-12 and the extension begins in 2014-01, so no revised datum and no later observation can change the replication label of Section 6.

8.1 Separating revision from the passage of time

An extension must use the current factor vintage while the baseline uses the publication-era vintage, so a naive comparison would confound a change over time with a change in data vintage. We therefore estimate a third system on the *baseline months* using the *current* vintage. Differencing it against the locked baseline isolates the vintage contribution; differencing the refitted extension against it isolates the temporal contribution. The registered gates are evaluated on the second comparison, so no revised historical datum enters the temporal verdict. A fourth system applies the 2013-12 parameters unchanged to the later months without re-estimating anything, and is genuinely out of sample.

Table 8: Temporal evaluations. The locked baseline and revised history share months and differ in vintage; the revised history and refitted extension share vintage and differ in months, so differencing isolates one change at a time.

Evaluation	Months	λ_{RM}	λ_{rate}	RMSE	Fit
Locked baseline, 1972–2013	504	0.6012	−0.6985	0.1004	0.544
Revised history, same months	504	0.6044	−0.6974	0.1015	0.534
Frozen parameters on 2014–2025	144	0.6012	−0.6985	0.4463	−0.691
Refitted extension, 2014–2025	144	0.9972	−0.0825	0.1918	0.450

The separation is clean. Recomputing the baseline window on a vintage nine years later moves the rate price of risk by 0.0011 and cross-sectional RMSE by 1.1 percent. Moving to the post-publication window moves them by 0.6148 and 89.0 percent. On every dimension the temporal contribution is one to two orders of magnitude larger than the vintage contribution, so the post-2013 change cannot be attributed to revised historical data.

8.2 All three registered gates fail

Sign compatibility of the rate-attributable fitted-premium spreads holds for two of seven families; the magnitude gate holds for one of seven; and RMSE deteriorates by 89.0 percent against a bound of 10 percent. The registered classification is *unsupported*. Five of the seven families reverse the sign of their spread, and the two that keep it do so with spreads that have collapsed toward zero. Applying the 2013-12 model unchanged to the later months gives a fit statistic of −0.691, meaning the fitted premia explain less of the cross-section of average returns than a constant would.

8.3 The obvious benign explanation does not hold

The extension window contains the effective lower bound, so the short-rate factor has far less variation in it: the innovation standard deviation falls from 0.5855 to 0.1707, and the share of months with the rate below half a percent rises from 12.3 to 41.7 percent. That invites the reading that the extension fails because the factor is no longer identified rather than because the relation changed.

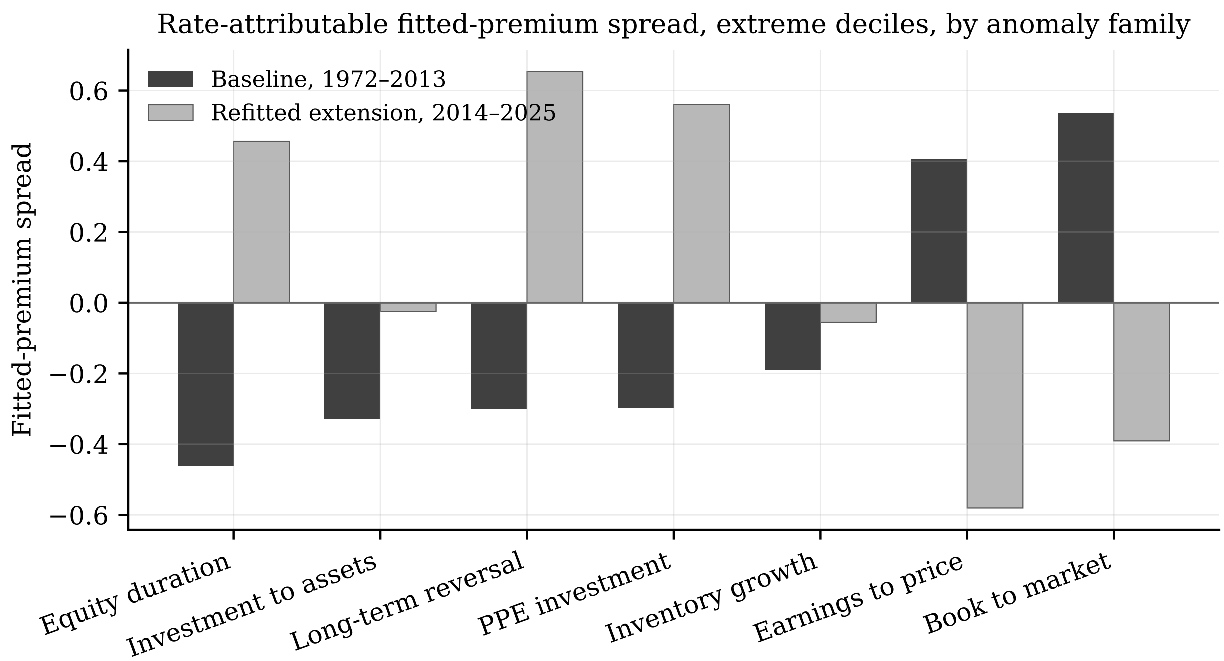

Figure 2: Rate-attributable fitted-premium spread between extreme deciles, by anomaly family, in the baseline window and in the refitted extension. Five of the seven families reverse sign; the two that keep it do so with spreads that have collapsed toward zero.

Source: `artifacts/tables/extension/fitted_premium_spreads.csv`.

The registered identification gate is not the factor's own variance but the cross-sectional dispersion of standardized exposures, which is scale aware by construction. That statistic *rises* from 0.2540 in the baseline to 0.5800 in the extension, against a floor of 0.10. Smaller factor variation is offset by larger betas, and the extension window clears the identification floor more comfortably than the baseline does.

The extension estimate is nonetheless imprecise in its own right, at -0.0825 with a Shanken t of -1.49 against -0.6985 and -2.86 in the baseline. A collapsed point estimate with a wide interval is consistent both with a relation that has weakened and with one estimated on 144 months instead of 504, and we do not choose between those.

The temporal analysis uses the extension-compatible monthly panel, frozen and refitted versions of the registered two-factor system, the same source-compatible anomaly portfolios in every window, and the predeclared temporal comparisons. Uncertainty is the Shanken covariance with the registered identification diagnostics; the economic magnitude is an 89.0 percent deterioration in cross-sectional RMSE against a 10 percent bound.

9 Monetary-regime stability

Pooled time-series interactions are evidence about exposure stability, not by themselves evidence about cross-sectional risk-price stability, which requires regime-specific second-pass models. The registered estimator therefore has two halves. They point the same way, but only one of them has the power to demonstrate it.

Two registered regimes straddle the 2013-12 vintage boundary, so a panel that switched vintage at its natural break would confound a policy difference with a data revision inside a single regime. The regime panel is built on the current vintage throughout, spanning 648 months from 1972-01 to 2025-12, which Section 8 licenses: over the baseline window the vintage contribution is 1.1 percent of RMSE against temporal contributions of 89.0 percent.

9.1 Most regimes cannot support a standalone second pass

The eligibility floors were fixed before any regime was estimated. Table 9 applies them. Only two of the six registered regimes clear the standalone second-pass tier, one of them limited to first-pass betas by the short-sample band and the remaining three falling below the first-pass floor entirely.

Table 9: Registered monetary regimes and their estimation eligibility under the frozen floors. Regimes without a standalone second pass enter the analysis only through the pooled interaction model.

Regime	Window	Months	Second pass	First pass
`conventional_pre_elb`	1972-01 to 2008-12	444	yes	yes
`elb_qe`	2009-01 to 2015-12	84	yes	yes
`normalisation`	2016-01 to 2020-03	51	no	yes
`pandemic_elb_qe`	2020-04 to 2022-03	24	no	no
`inflation_tightening`	2022-04 to 2024-09	30	no	no
`post_tightening_easing`	2024-10 to 2025-12	15	no	no

The confirmatory equivalence comparison therefore spans a single contrast. The four ineligible regimes are not dropped: they enter through the pooled interaction model, which borrows strength across the whole sample.

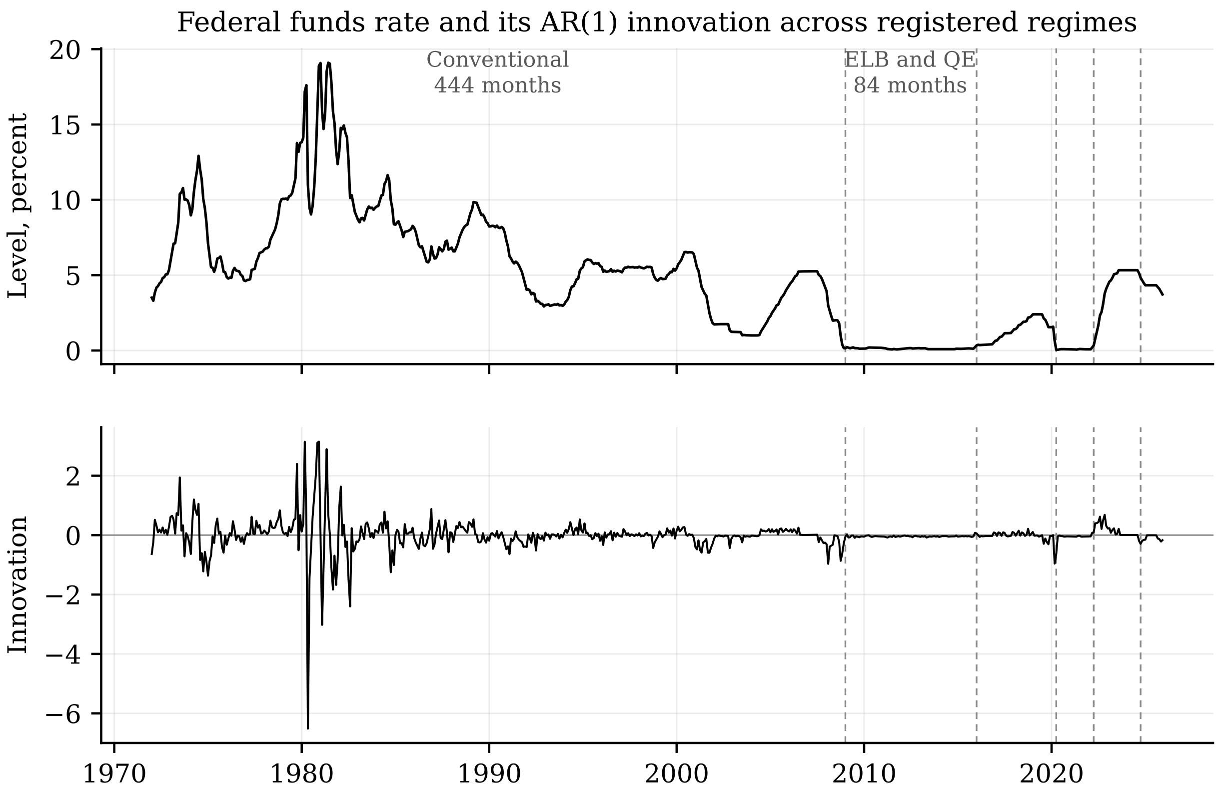

Figure 3: The federal funds rate and its autoregressive innovation across the registered regimes, on the vintage-consistent panel. Dashed lines mark regime boundaries. The innovation that identifies the factor retains almost none of its earlier variation once the policy rate reaches the effective lower bound.

Sources: `artifacts/tables/regimes/regime_monthly_series.csv` and `artifacts/tables/regimes/regime_eligibility.csv`.

9.2 The two estimable regimes

Table 10: Regime-specific second passes on the joint seventy-portfolio system, current vintage throughout. Only regimes clearing the frozen eligibility floors appear. Dispersion is the cross-sectional standard deviation of rate-attributable fitted premia.

Regime	Months	λ_{RM}	λ_{rate}	Shanken t	RMSE	Fit	Dispersion
Conventional, 1972–2008	444	0.458	−0.7259	−2.87	0.1046	0.596	0.1753
Lower bound, 2009–2015	84	1.327	0.0037	0.84	0.2220	−0.172	0.0333

The conventional-regime estimate reproduces the baseline closely, at −0.7259 on 444 months against −0.6985 on the 504-month baseline, with the same Shanken t-ratio to two decimals. The lower-bound estimate is a different object: the price of risk is not merely smaller but indistinguishable from zero, and the fit statistic is negative. The average absolute rate-attributable fitted premium falls from 0.1517 to 0.0281 monthly percentage points, and every one of the seven families moves toward zero rather than reversing.

9.3 Equivalence is not demonstrated, and neither is difference

The confirmatory rule is two one-sided tests at the 5 percent level, implemented as inclusion of the two-sided 90 percent joint-bootstrap interval inside the declared bound, applied to the rate-attributable fitted premium of every one of the seventy test assets. All five registered gates fail to certify equivalence.

An equivalence test can fail two ways, and only one of them is a finding. The interval can lie wholly beyond the bound, which is evidence of a real difference, or it can straddle the bound, which means the data cannot resolve the question. Sorting the seventy per-portfolio tests: 26 certify equivalence, 44 are inconclusive, and *none* demonstrates a change exceeding the registered bound of 0.25 monthly percentage points. Sixteen have point estimates outside the bound, the largest being 0.5272, but in every case the interval reaches back inside it.

Among the aggregate gates the pattern is the same but for one exception. The relative deterioration in cross-sectional RMSE has its entire 90 percent interval above the bound, at $[+11.4, +135.9]$ percent against 10 percent; the dispersion, maximum pricing error and fit comparisons are all inconclusive.

The registered outcome is *unsupported*, and the wording matters. That label asserts regime dependence, which is a positive claim, so we assign it only because one dimension demonstrates an exceedance. Had that gate also been inconclusive the registered outcome would have been the inconclusive one, since no other equivalence dimension supplies affirmative evidence.

9.4 The pooled model settles the direction

Using all 648 months and interacting both factors with the regime label, the restriction that rate betas are equal across regimes is rejected with a statistic of 37.31 on 5 degrees of freedom, an unadjusted p-value of 5.2×10^{-7} and a Holm-adjusted p-value of 5.1×10^{-5}; the all-factor restriction is rejected more strongly still. Per asset, rate-beta interactions are significant for 26 of 70 after adjustment. Holm was applied across the 142 pooled tests, and the registered family also contains the equivalence tests above; a Bonferroni bound over the completed family of 216 leaves both joint results significant, so the conclusion does not depend on where the family boundary is drawn.

One fragility is recorded. Shifting every regime boundary by three months either way leaves the verdict intact but not the aggregate statistic that produces it: its adjusted p-value rises to 0.065 and 0.051 respectively, so the registered boundaries are the most favourable of the three for that statistic. The verdict survives on the per-asset rejections, which increase to 39 and 36.

A battery of structural-break tests is reported as exploratory and is excluded from the confirmatory family. Its most useful result is negative: left to find breaks on its own, the sample selects 1989-12, 1998-07 and 2001-08, none of them a monetary-policy date, and formal tests at the registered boundaries do not reject at the 5 percent level. The dates on which this cross-section breaks are not the dates on which monetary policy changed.

9.5 Why the equivalence tests could not resolve the question

A simulation calibrated to the estimated process, with 2,000 replications at each of sixteen window lengths, shows the inconclusive verdicts are a property of the estimator at these sample sizes. On the joint seventy-asset system, resolving fitted-premium spreads against the 0.25 bound requires 180 months; the lower-bound regime has 84. Driving the probability of a sign error on the rate price of risk below 5 percent requires 444 months.

The mechanism is errors-in-variables attenuation, and it does not wash out with sample size: even at 648 months, 66.7 percent of the cross-sectional dispersion of estimated rate betas is first-pass sampling noise, against 2.9 percent for market betas. The sampling standard deviation of the rate price of risk is essentially flat, 0.1499 at 12 months and 0.1167 at 648, so a longer window moves the estimate rather than tightening it. The market price of risk is the control that shows the machinery is sound, with coverage between 0.943 and 0.959 at every window length. Because the calibration treats estimated betas as true, these levels are comparable across window lengths rather than interpretable as absolute measurements, and the simulated cross-section is if anything better identified than the real one.

The regime analysis uses the vintage-consistent 648-month panel, the predeclared regime indicators, pooled interaction tests for exposures and separate second-pass models for prices of risk, and the same seventy anomaly portfolios throughout. Uncertainty comes from Holm-adjusted Wald tests and the joint moving-block bootstrap; the economic magnitude is a fall in the average absolute rate-attributable fitted premium from 0.1517 to 0.0281 monthly percentage points between the two estimable regimes.

10 Discussion and limitations

The first limitation is the input layer, and it bounds every claim in the paper. The article names its providers without series codes or archive vintages, and the anomaly decile panels come from the original author's source but in a later rebuild that does not match the published descriptive statistics on any of the seven families. No result here is therefore eligible for an exact-replication label at any recovery rate. The consequence is asymmetric: the reconstruction can show that a published pattern reappears, and it can show that a pattern fails to extend, but it cannot attribute a cell-level discrepancy to the article rather than to the inputs. Where the audit falls short of published precision we classify the layer as partially recovered and say nothing stronger.

The second limitation is one of our own design. The registered materiality standard mixes two relative gates set at 10 percent with an absolute gate of 0.25 monthly percentage points, and on these test assets that absolute gate is equivalent to demanding a 52.6 percent reduction. The mixture makes the third gate roughly five times stricter than the other two, which we do not believe was intended, and it alone determines the headline classification. The threshold was frozen before any estimate existed and is applied as frozen; a future pre-registration should express the three gates on a common scale.

The third limitation is econometric and is the binding one for the regime evidence. Cross-sectional pricing is fragile when betas carry a large share of sampling noise, and the simulation in Section 9 shows that share is 66.7 percent for rate betas even at 648 months. Regime-length samples cannot resolve fitted-premium differences against the registered bound, which is why 44 of 70 equivalence tests are inconclusive. This is a statement about what the estimator can support at those sample sizes, and it applies to any study that partitions a monthly cross-section into policy regimes.

The fourth limitation is interpretive. A priced covariance with a rate innovation would not by itself establish a narrow monetary transmission mechanism. Our regime evidence sharpens the point in an unwelcome direction: the exploratory break tests locate their breaks at dates that are not policy dates. We therefore separate pricing evidence from identification evidence throughout and keep the high-frequency policy-information decomposition as an appendix design until usable event data exist.

11 Conclusion

Rebuilding Maio and Santa-Clara (2017) from independently acquired sources, we recover the published qualitative result on the original window: short-rate innovations carry a negative and statistically significant price of risk in the cross section of anomaly portfolios, and adding the factor to the market model roughly halves cross-sectional pricing errors.

The extensions are where the paper contributes, and both are negative. The short-rate factor does not meet the incremental-pricing standard we registered in advance, and on the same seventy portfolios every traded multi-factor comparator prices more accurately than it does. The relation then does not survive its own publication sample: refitted post-2013 estimates reverse sign for five of seven anomaly families and deteriorate far beyond the registered bound. We can rule out two benign readings of that failure, since the change is an order of magnitude larger than the data-vintage contribution and the factor clears the registered identification gate more comfortably after 2013 than before.

Across monetary regimes the evidence is asymmetric by construction. Exposures are demonstrably regime dependent under the pooled test, while the size of the change in fitted premia cannot be certified at feasible regime lengths in either direction. We report that asymmetry rather than resolving it, because the simulation shows the imprecision is a property of two-pass estimation on regime-length samples rather than a fact about the periods.

What survives is narrower than the original claim and, we think, more useful: a short-rate innovation is well identified and was priced in the cross-section of anomaly returns through 2013, it was never the most accurate available pricing model on those assets, and it has not been priced since.

A Table-level replication audit

The table-level audit maps the article and supplement targets to replication statuses. It covers the article tables and the supplement appendix tables through Table A.14. Cell-level comparison was run against the statistics transcribed from Tables 3, 4, 6 and A.1, which after collapsing rows that repeat a point estimate under two printed uncertainty measures give 123 unique published cells. All 123 were compared.

Table 11: Cell-level recovery by statistic. A cell counts as recovered when it agrees with the article to the precision the article prints, that is within half of the last printed increment.

Statistic	Cells	Recovered	Share	Median $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$
Market risk price	29	20	0.690	0.0033	0.0117
Rate risk price	16	3	0.188	0.0137	0.0901
Cross-sectional fit	29	3	0.103	0.0251	0.2863
Specification statistic	29	0	0.000	0.7458	7.4110
Constrained fit	5	1	0.200	0.0162	0.0492
Comparator factor prices	15	2	0.133	0.0193	0.1350

Table 12: Replication layer classification. No layer is eligible for an exact-replication label, because no input is an exact article input.

Layer	Recovered	Classification
R1a short-rate innovations	per series	approximately reproduced under documented reconstruction
R1b first-pass betas	—	no published statistic level target
R1c risk prices	23 of 45	partially recovered under documented reconstruction
R1d pricing errors and fit	3 of 58	partially recovered under documented reconstruction
R1e comparator models	3 of 20	partially recovered under documented reconstruction

R1b carries no statistic-level target. The article plots first-pass betas and reports beta-times-lambda decompositions for the extreme deciles, but tabulates no beta, so there is nothing to compare cell by cell and the layer is evidenced only through the layers that consume it. This is a property of what the article published, not a gap in the reconstruction.

Three groups of targets were not attempted and are recorded as such rather than being compared against a different object. Every bootstrap p-value in the published tables comes from the article's own useless-factor bootstrap, which is not implemented here. Table 5, Tables 7 to 9 and Appendix Tables A.2 to A.14 lie outside this pass. Equal-weighted results are blocked at the current sources.

Table 13: Original table groups and registered targets

Original evidence group	Registered targets	Status this pass
Article descriptive and pricing tables	TBL_001–TBL_009	Compared where a statistic-level target exists
Supplement interest-rate robustness	APP_TBL_A01–APP_TBL_A06	Partially compared
Supplement portfolio and model extensions	APP_TBL_A07–APP_TBL_A14	Not attempted this pass

Source: `research/table_target_manifest.csv`.

B Data access and reproducibility

Restricted article and supplement files remain local-only. The committed repository records metadata and source-access status, but it does not redistribute copyrighted source documents, prompt files, licensed data extracts, local catalogs, temporary artifacts, or machine-specific environment manifests. The source-only release gate reports zero critical restricted-path issues and one major missing-input issue.

Source-access requirements, release artifacts, and validation status are recorded outside the main argument in the repository documentation and generated release manifests.

C Hypothesis registry

The confirmatory registry contains a replication target and hypotheses for incremental pricing content, post-publication compatibility, regime equivalence, and weak-factor strength. The registry distinguishes null, alternative, primary outcome, estimator, sample, test assets, economic direction or equivalence criterion, multiplicity family, and supported, unsupported, or inconclusive interpretation. Exploratory structural-break alignment is recorded separately from confirmatory claims.

D Table and figure architecture

Every result table and figure in this paper is generated from the frozen artifacts rather than transcribed. `scripts/build_manuscript_tables.py` writes the eight result tables and `scripts/build_manuscript_figures.py` the two figures; the manuscript includes the emitted files, and each generated row carries a comment naming the artifact it came from. A test regenerates every table and fails if the committed file differs, so a regenerated artifact cannot leave a stale number in the paper.

The registered architecture also anticipated figures for beta-return relations, rolling risk prices, and regime equivalence intervals. Those are not produced here. The equivalence intervals in particular would invite reading a per-portfolio contrast that Section 9 shows the sample cannot resolve, and we prefer to state that limitation in text than to draw it.

E Detailed out-of-sample design

The out-of-sample design freezes an initial training endpoint at 1999-12 and uses a predeclared annual refit schedule. Forecast records must store model vintage, training dates, factor definition, asset universe, benchmark model, and loss function. Results from this design will be interpreted as evidence about stability of pricing errors, not as a substitute for the baseline replication audit.

F Optional policy-information decomposition

The policy-information decomposition is not part of the main contribution until legally usable event data, component labels, monthly aggregation artifacts, and component-level weak-factor diagnostics are available. If included, the components are high-frequency components constructed under an explicit announcement-based identification design, not automatically identified objects. The monthly aggregation can create sparse factors with many no-event months, so the decomposition must pass factor-strength diagnostics before entering the main pricing argument.

G Additional diagnostics

Additional diagnostics include standardized exposure dispersion, rank checks, factor spanning, influence diagnostics, leave-one-family systems, boundary-shift checks, and multiple-testing adjustments. Raw beta dispersion, singular values, and condition numbers are descriptive diagnostics only. Diagnostic outputs remain pending until the relevant baseline and extension artifacts exist.

References

Bai, J. and Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78.

Bernanke, B. S. and Kuttner, K. N. (2005). What explains the stock market's reaction to federal reserve policy. *Journal of Finance*, 60(3):1221–1257.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636.

French, K. R. (2026). Kenneth r. french data library. `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`. Accessed 2026-07-31.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *The Review of Financial Studies*, 28(3):650–705.

Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises: The role of information shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Kan, R. and Zhang, C. (1999). Two-pass tests of asset pricing models with useless factors. *Journal of Finance*, 54(1):203–235.

Lewellen, J., Nagel, S., and Shanken, J. (2010). A skeptical appraisal of asset-pricing tests. *Journal of Financial Economics*, 96(2):175–194.

Lioui, A. and Maio, P. (2014). Interest rate risk and the cross section of stock returns. *Journal of Financial and Quantitative Analysis*, 49(2):483–511.

Maio, P. F. and Santa-Clara, P. (2017). Short-term interest rates and stock market anomalies. *Journal of Financial and Quantitative Analysis*, 52(3):927–961.

Merton, R. C. (1973). An intertemporal capital asset pricing model. *Econometrica*, 41(5):867–887.

Newey, W. K. and West, K. D. (1987). A simple positive semi-definite heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Shanken, J. (1992). On the estimation of beta-pricing models. *The Review of Financial Studies*, 5(1):1–33.

Swanson, E. T. and Williams, J. C. (2014). Measuring the effect of the zero lower bound on medium- and longer-term interest rates. *American Economic Review*, 104(10):3154–3185.